A Weighted Coherence Estimator for SAR Coherent Change Detection

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***Abstract*— Synthetic aperture radar (SAR) coherent change detection (CCD) predominantly uses the degree of coherence or similar change metrics as a measurement of the changes that have occurred between two data collections. Many existing coherence estimators have shown some performances for change detection, but are still relatively limited, because the change areas do not stand out well from all decorrelation areas due to low clutter-to-noise ratio (CNR) and volume scattering. Besides, many estimators require the equal-variance assumption between two SAR images of the same scene. However, the assumption is less likely to be met in areas with significant intensity differences, especially in change regions. In this article, a novel weighted coherence estimator is proposed to address these problems. The estimator is derived based on the statistical characteristics of SAR images using the maximum-likelihood (ML) principle. The introduction of weight parameters not only makes the estimator no longer need to satisfy the equal-variance assumption but also combines the advantages of coherent and noncoherent algorithms to a certain extent, because the weights are closely related to the ratio change statistic. Experiments on simulated and real SAR image pairs demonstrate the effectiveness of the proposed estimator in highlighting change areas and the boundaries between change and other areas in theory and practice.**

***Index Terms*— Coherent change detection (CCD), synthetic aperture radar (SAR), weighted coherence estimator.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a coherent imaging technique with the capability of generating high-resolution images of the complex radar backscatter of a scene. As an important application of SAR, SAR change detection utilizes two SAR images of the same scene taken at different times to detect changes that have occurred between the two acquisition dates [1].

Two forms of traditional SAR change detection are generally considered, that is, coherent and incoherent change detection. Incoherent change detection uses SAR intensity (amplitude squared) images to identify large-scale changes [2]. Coherent change detection (CCD), on the other hand, estimates the coherence between the two SAR images using SAR phase and intensity to quantify changes. As mentioned in [2]–[5], CCD is capable of detecting subtle changes, indicating the ability to keep more change details than incoherent intensity change detection algorithms.

As coherence statistics generally provide low coherence values in change areas, decorrelation is a key problem for CCD. CCD seeks to distinguish decorrelation due to change from one due to other reasons. In practice, there are many sources of decorrelation, such as baseline or geometric decorrelation, Doppler centroid decorrelation, processing induced decorrelation, thermal or system noise, volume decorrelation, and temporal terrain decorrelation [6]. For a given pair of spatial-registered SAR images, the sources of decorrelation to be considered mainly contain temporal decorrelation, volume decorrelation, and low clutter-to-noise ratio (CNR) decorrelation. Among them, temporal decorrelation is caused by physical changes in the terrain [6]. Obviously, changes that have occurred between the first and the second data acquisition, the interest of change detection, fall into this decorrelation category. Volume decorrelation is caused by penetration of the radar wave in the scattering medium [6], [7]. Also, vegetation terrain, e.g., forest terrain, is one typical medium with significant radio wave volume scattering. Besides, vegetation growth inevitably results in temporal decorrelation, which is often a disturbance for most change detection tasks except ones of vegetation. The last low-CNR decorrelation can occur for many features in the SAR scene, including shadow areas, hard-packed roads, smooth water surfaces or other smooth surfaces, and materials with low radar reflectivity [8]. It is well known that the purpose of CCD is to detect changes that have occurred between two SAR images; therefore, the capability of coherence estimators in identifying change areas from all decorrelation areas has an important influence on the accuracy of the detection results.

At present, many coherence estimators have been proposed. At the early time, the coherence statistic [9], [10] is used for terrain height measures for repeat-pass interferometric SAR

Manuscript received February 20, 2022; revised May 11, 2022; accepted May 23, 2022. Date of publication June 3, 2022; date of current version June 14, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 42171442; in part by the Open Research Fund of Jiangsu Key Laboratory of Resources and Environmental Information Engineering, China University of Mining and Technology, under Grant JS202106; and in part by the National Natural Science Foundation of China under Grant 51774270. *(Corresponding author: Lijun Lu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3180044

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image pairs. Jakowatz *et al.* [11] first proposed SAR CCD to detect anthropogenic temporal changes using a 1-m resolution spotlight interferometric image pair. To overcome decorrelation measurements due to low CNR in the images, Yocky and Johnson [12] suggested masking the potential low-CNR areas using the CNR information from the SAR image pair. Other authors derived a new maximum-likelihood (ML) change estimator, the complex reflectance change detection (CRCD) metric, to distinguish change regions from low-CNR areas [8]. Similarly, some log-likelihood change statistics were proposed based on a Bayesian hypothesis-testing framework [13], [14]. Details and limitations of these algorithms will be presented later in Section II. In addition, some change methods combine coherent and noncoherent algorithms, using intensity and coherence characteristics of SAR images, for SAR change detection [2], [15]. Here, these methods can be seen as a combination of two types of methods, and coherent approaches always utilize the abovementioned coherent estimators. Performances of these algorithms are still relatively limited, because the change areas do not stand out well from all decorrelation areas due to low CNR and volume scattering. In particular, the boundaries between change and no-change regions are fuzzy and difficult to determine, unfavorable for obtaining fine change maps.

Furthermore, most coherence estimators are employed to detect changes in high-resolution images, such as airborne SAR images. In practice, satellite SAR images seem to have a wider application than airborne images, such as TerraSAR/TanDEM images, but the resolution of these types of images is generally lower than airborne images except for some special imaging modes (IMs). As we know, a resolution cell is the coherent sum of a large number of scattering contributions [16], and a lower resolution cell often contains more scatters. It is worth investigating the applicability of the coherence estimators in satellite SAR images with relatively lower resolution than airborne images.

To address the abovementioned problems, we propose a new weighted coherence estimator to better highlight change regions and investigate its applicability in satellite images. The estimator is derived based on the statistical characteristics of SAR images using the ML principle. As the weighted parameters are closely related to the ratio change statistic used in noncoherent algorithms, its introduction into the derivation makes the estimator combine the advantages of coherent and noncoherent algorithms to a certain extent, which is demonstrated in real satellite SAR images in Section V. Section II presents some classical coherence estimators for CCD and some limitations of these estimators. Section III first introduces the statistical characteristics of SAR images and then gives the derivation of the weighted estimator. Experiments on simulated and real satellite SAR image pairs are presented in Sections IV and V, respectively, validating the effectiveness of the new estimator in highlighting changes and the boundaries between change and other areas in theory and practice. Finally, Section VI concludes this article.

II. Limitations of the Coherence Estimator and Its Modifications

CCD is applicable to complex data and utilizes both the magnitude and phase of SAR images. As a change metric of CCD, the coherence γ is often estimated using a classical coherence estimator [9], [10]

$$\hat{\gamma}_c = \frac{\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sqrt{\sum_{k=1}^{N} f_k^* f_k \sum_{k=1}^{N} g_k^* g_k}} \tag{1}$$

where $\boldsymbol{X} = [\boldsymbol{f}, \boldsymbol{g}]^T \in \mathbb{C}^{N\times 2}$ is a jointly circular, zero-mean, Gaussian random vector and consists of N sample pairs of observations $\boldsymbol{X}_k = [f_k, g_k], k = 1, 2, \ldots, N$, and the symbol "$*$" is the complex conjugate operation. It should be mentioned that underlying parameters and their estimator are differentiated by placing a hat $\hat{}$ over the latter. More modifications can be seen in [17] and [18].

The metric $\hat{\gamma}_c$ has been widely used in interferometric SAR applications for terrain height generation, as it is essentially a measure of the similarity degree in the image pair. Thus, this coherence measure will provide low values in low-similarity areas, including change, low CNR, and many vegetation areas, which will greatly decrease change detection performance. An alternative coherence estimator, derived by Berger [19], is then explored as a means of CCD and demonstrated better than $\hat{\gamma}_c$ for change detection in a certain condition [2]. This complex correlation coefficient is given by

$$\hat{\gamma}_b = \frac{2\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sum_{k=1}^{N} f_k^* f_k + \sum_{k=1}^{N} g_k^* g_k}. \tag{2}$$

The alternative metric assumes variance equality and is a more stable estimator using a sum rather than a product of two random variables in the denominator. To better determine if the true change has actually occurred in low-CNR regions, several methods have been proposed in [8] and [12]–[14]. Typically, Wahl *et al.* [8] introduced a change measure similar to the coherence and added a system thermal noise term for each observation. Also, the CRCD metric is expressed as

$$\hat{\gamma}_a = \frac{2\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sum_{k=1}^{N} f_k^* f_k + \sum_{k=1}^{N} g_k^* g_k - N\sigma_{nf}^2 - N\sigma_{ng}^2} \tag{3}$$

where σ_{nf}^2 and σ_{ng}^2 demote the variances of the additive noise for two random variables $\boldsymbol{f}$ and $\boldsymbol{g}$, respectively.

Although these estimators show good performance for change detection, the assumption of equal variance between two random variables should be met for most estimators. As [20] has investigated, the variance of a given pixel in an SAR image is determined by the scattering amplitudes and is closely related to the average intensity. As we know, there are often significant intensity differences in change regions, which makes the equal variance assumption less likely to be met in these areas. Consequently, the change information may not be nicely highlighted in the image generated from the abovementioned coherence estimators. In this article, a weighted ML coherence estimator for CCD has been proposed to address the problem.

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III. Methodology

A. Statistical Description of SAR Image Pair

The SAR data are known as the complex image, which is consisted of two parts: images of the real part $A\cos\Phi$ and the imaginary part $A\sin\Phi$, where A is the amplitude, and Φ is the phase. Also, the observed real and imaginary components, $z_1 = A\cos\Phi$ and $z_2 = A\sin\Phi$, are often viewed as independent variables drawn from an identical zero-mean Gaussian distribution and whose variance $\sigma^2/2$ is determined by the scattering amplitude [20]. In addition, they will have a joint probability density function (pdf) given by

$$p(z_1, z_2) = \frac{1}{\pi\sigma^2}\exp\left(-\frac{z_1^2 + z_2^2}{\sigma^2}\right) \tag{4}$$

where σ^2 corresponds to the average intensity. Here, the word "intensity" is by analogy with measurements at optical wavelengths and can be described as $I = A^2$.

Given two spatial registered SAR images, a joint data vector $\boldsymbol{X} = [\boldsymbol{f}, \boldsymbol{g}]^T \in \mathbb{C}^{N\times 2}$ can be formed, the same definition as mentioned in Section II. Its conditional density function can be expressed as

$$p(\boldsymbol{X}|\gamma, \phi) = \frac{1}{\pi^{2N}|\boldsymbol{Q}|^N}\exp\left(-\sum_{k=1}^{N}\boldsymbol{X}_k^H\boldsymbol{Q}^{-1}\boldsymbol{X}_k\right) \tag{5}$$

where the superscript H is the complex conjugate transpose operation, and $|\boldsymbol{Q}|$ is the determinant of the covariance matrix $\boldsymbol{Q}$ given by

$$\boldsymbol{Q} = \mathbb{E}(\boldsymbol{X}\boldsymbol{X}^H) = \begin{bmatrix} \sigma_f^2 & \gamma\sigma_f\sigma_g e^{j\phi} \\ \gamma\sigma_f\sigma_g e^{-j\phi} & \sigma_g^2 \end{bmatrix} \tag{6}$$

where $\sigma_f^2 = \mathbb{E}(|\boldsymbol{f}|^2)$, $\sigma_g^2 = \mathbb{E}(|\boldsymbol{g}|^2)$, γ is the magnitude of the complex cross-correlation coefficient, and ϕ denotes the phase difference that exists between $\boldsymbol{f}$ and $\boldsymbol{g}$.

B. Proposed Weighted Coherence Estimator

As an unknown parameter, the coherence magnitude γ can be estimated by maximizing (5) or, equivalently, the log of (5), based on the ML principle. The new estimate is found by taking the partial derivative concerning γ of the log of (5), setting the result to zero and solving for γ.

The log of (5) is easily gotten as

$$\ln p(\boldsymbol{X}|\gamma, \phi) = -2N\ln\pi - N\ln|\boldsymbol{Q}| - \sum_{k=1}^{N}\boldsymbol{X}_k^H\boldsymbol{Q}^{-1}\boldsymbol{X}_k. \tag{7}$$

To simply the notation, the summation symbol $\sum_{k=1}^{N}$ is replaced with $\sum$ in the rest of this article. Then, taking the partial derivative of (7) concerning γ and setting the result to zero give

$$-\frac{N}{|\boldsymbol{Q}|}\frac{\partial(|\boldsymbol{Q}|)}{\partial\gamma} - \sum\boldsymbol{X}_{\boldsymbol{k}}^H\frac{\partial(\boldsymbol{Q}^{-1})}{\partial\gamma}\boldsymbol{X}_{\boldsymbol{k}} = 0 \tag{8}$$

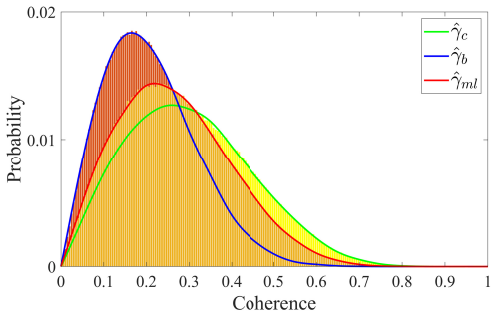

Fig. 1. Fitting probability curves based on the histograms of the coherence images calculated from different coherence estimators.

where $|\boldsymbol{Q}|$ is the determinant of $\boldsymbol{Q}$, and $\boldsymbol{Q}^{-1}$ is the inverse matrix defined by

$$|\boldsymbol{Q}| = (1-\gamma^2)\sigma_f^2\sigma_g^2 \tag{9}$$

$$\boldsymbol{Q}^{-1} = \frac{1}{1-\gamma^2}\begin{bmatrix} \frac{1}{\sigma_f^2} & -\frac{\gamma e^{j\phi}}{\sigma_f\sigma_g} \\ -\frac{\gamma e^{-j\phi}}{\sigma_f\sigma_g} & \frac{1}{\sigma_g^2} \end{bmatrix}. \tag{10}$$

Substituting the abovementioned definition for $|\boldsymbol{Q}|$ and $\boldsymbol{Q}^{-1}$ into (8) and replacing ϕ with its ML estimate (see [9]) yield the following equation after a large amount of algebraic manipulation:

$$N\gamma(1-\gamma_2) - \gamma\left(\frac{\sum f_k^* f_k}{\sigma_f^2} + \frac{\sum g_k^* g_k}{\sigma_g^2}\right) + \frac{1+\gamma_2}{\sigma_f\sigma_g}\left|\sum f_k^* g_k\right| = 0. \tag{11}$$

Substituting the following reasonable approximations:

$$\frac{1}{N}\sum f_k^* f_k \approx \sigma_f^2 \tag{12}$$

$$\frac{1}{N}\sum g_k^* g_k \approx \sigma_g^2 \tag{13}$$

into (11), we get the only viable solution as (see the Appendix for a detailed derivation)

$$\hat{\gamma}_{ml} = \frac{2\left|\sum f_k^* g_k\right|}{\sum f_k^* f_k\Big/\sqrt{R} + \sqrt{R}\sum g_k^* g_k} \tag{14}$$

where $R = \sigma_f^2/\sigma_g^2$ is the ratio of the true variance.

The coherence metric $\hat{\gamma}_{ml}$ can be viewed as a weighted form of the estimator $\hat{\gamma}_b$. To compare the effectiveness of different estimators, here, we randomly simulated two single look complex (SLC) SAR images of the size of 1000 × 1000 based on the assumption described in Section III-A. The variance σ_g^2 and the ratio R are set to 1.0 and 0.5, respectively. Fig. 1 shows the probability curves fitting by observations, i.e., the histograms of the coherence images calculated from different coherence estimators. It is well known that the coherence metrics will provide a low value in regions where the change has occurred. As a random simulation, the coherence values of two SLC images are generally very low (see Fig. 1). Besides, the ratio R is set

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to be a value much smaller than 1. Therefore, the simulation images based on the test $\sigma_f \neq \sigma_g$ and $\gamma < 1$ can be viewed as ones covering a scene of change (for detailed explanation, see Section IV). To better highlight the change, the smaller the better as far as the coherence magnitude estimated using the estimator in the change area. However, notice from Fig. 1 that the weighted estimator $\hat{\gamma}_{ml}$ seems more likely a compromise between estimators $\hat{\gamma}_b$ and $\hat{\gamma}_c$, which is not what we expect in CCD.

It is also true, however, that the weighting operation is effective to improve the coherence magnitude compared with $\hat{\gamma}_b$. In CCD, we expect to lower the coherence magnitude in change regions to improve the contrast between the changed areas and the background (unchanged areas). Inspired by the idea, the new weighted estimator is defined as

$$\hat{\gamma}_w = \frac{2\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sqrt{R}\sum_{k=1}^{N} f_k^* f_k + \sum_{k=1}^{N} g_k^* g_k \Big/ \sqrt{R}}. \tag{15}$$

Note that the weights of $\hat{\gamma}_w$ are replaced with the reciprocals of that of $\hat{\gamma}_{ml}$ to decrease the coherence in change areas. To demonstrate the rationality of the abovementioned definition, we will prove it in two cases: $R < 1$ and $R \cong 1$. First, we can get the following relationship using the definition of R and the approximations in (12) and (13) as:

$$\sum f_k^* f_k \approx R \sum g_k^* g_k. \tag{16}$$

Then, the denominator of (15) can be expressed as

$$\sqrt{R}\sum f_k^* f_k + \sum g_k^* g_k \Big/ \sqrt{R} \approx \frac{1+R^2}{\sqrt{R}} \sum g_k^* g_k. \tag{17}$$

Similarly, the denominator of $\hat{\gamma}_b$ can be written as

$$\sum f_k^* f_k + \sum g_k^* g_k \approx (1+R) \sum g_k^* g_k. \tag{18}$$

Thus, we get the difference Δ of these two denominators as

$$\begin{aligned} \Delta &= \left(\frac{1+R^2}{\sqrt{R}} - (1+R)\right) \sum g_k^* g_k \\ &= \frac{\left(1-\sqrt{R}\right)\left(1-R\sqrt{R}\right)}{\sqrt{R}} \sum g_k^* g_k. \end{aligned} \tag{19}$$

In case of $R < 1$, we can easily know that $\Delta > 0$, which means

$$\sqrt{R}\sum f_k^* f_k + \sum g_k^* g_k \Big/ \sqrt{R} > \sum f_k^* f_k + \sum g_k^* g_k. \tag{20}$$

Combining (20) and the inequality property, we can easily get

$$\begin{aligned} \frac{2\left|\sum f_k^* g_k\right|}{\sqrt{R}\sum f_k^* f_k + \sum g_k^* g_k \Big/ \sqrt{R}} &< \frac{2\left|\sum f_k^* g_k\right|}{\sum f_k^* f_k + \sum g_k^* g_k} \\ &< \frac{\left|\sum f_k^* g_k\right|}{\sqrt{\sum f_k^* f_k \sum g_k^* g_k}} \end{aligned} \tag{21}$$

that is,

$$\hat{\gamma}_w < \hat{\gamma}_b < \hat{\gamma}_c. \tag{22}$$

Under the assumption that the variance between the data $\boldsymbol{f}$ and $\boldsymbol{g}$ is almost equal, that is, $R \cong 1$ and

$$\left|\sum_{k=1}^{N} f_k^* f_k\right| \cong \left|\sum_{k=1}^{N} g_k^* g_k\right| \tag{23}$$

we have the following relationship:

$$\begin{aligned} \hat{\gamma}_w &\cong \frac{2\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sum_{k=1}^{N} f_k^* f_k + \sum_{k=1}^{N} g_k^* g_k} \cong \frac{2\left|\sum_{k=1}^{N} f_k^* g_k\right|}{2\sum_{k=1}^{N} f_k^* f_k} \\ &= \frac{\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sqrt{\left(\sum_{k=1}^{N} f_k^* f_k\right)^2}} \cong \frac{\left|\sum_{k=1}^{N} f_k^* g_k\right|}{\sqrt{\sum_{k=1}^{N} f_k^* f_k \sum_{k=1}^{N} g_k^* g_k}}. \end{aligned} \tag{24}$$

In summary, compared with $\hat{\gamma}_c$ and $\hat{\gamma}_b$, the proposed $\hat{\gamma}_w$ can theoretically give the lowest coherence values in the condition of $R < 1$ and provide approximate values when the equal variance assumption is met. One point to be mentioned here is that we do not consider the case of $R > 1$, because it is virtually the same as the case of $R < 1$, just exchanging the order of vectors $\boldsymbol{f}$ and $\boldsymbol{g}$. Also, the feasibility of the new estimator will be further verified on simulated and real SAR images in Sections IV and V, respectively.

IV. Simulated Data Example

To theoretically verify the feasibility of the proposed estimator, an SAR SLC image pair of the size of 2000×2000 is simulated. In the simulation process, we adopt the hypothesis tests in [2] as

$$\begin{aligned} &H_0^{1\text{st}} : \sigma_f \approx \sigma_g \quad H_1^{1\text{st}} : \sigma_f \neq \sigma_g \\ &H_0^{2\text{nd}} : \gamma \approx 1 \quad H_1^{2\text{nd}} : \gamma < 1. \end{aligned} \tag{25}$$

Among the two sets of hypothesis tests, there are four possible outcomes: $H_0^{1\text{st}} + H_0^{2\text{nd}}$, $H_0^{1\text{st}} + H_1^{2\text{nd}}$, $H_1^{1\text{st}} + H_0^{2\text{nd}}$, and $H_1^{1\text{st}} + H_1^{2\text{nd}}$. Similar to [2], we also consider that $H_1^{1\text{st}} + H_0^{2\text{nd}}$ is unlikely to happen. Therefore, the simulated images are only based on the other three hypothesis tests.

Hypothesis test $H_0^{1\text{st}} + H_0^{2\text{nd}}$ indicates that the final no change. Under this assumption, the simulated images can be viewed as a high-coherence image pair where no change has occurred. For this simulation image pair, the real and imaginary components of one image are first randomly simulated, and then added a Gaussian noise term for two components, respectively, as the other SLC image.

Relative to these images with high coherence, the simulated images relied on tests $H_0^{1\text{st}} + H_1^{2\text{nd}}$ and $H_1^{1\text{st}} + H_1^{2\text{nd}}$ are regarded as ones of low coherence. Therefore, the SLC images are generated using a random Gaussian complex simulation to keep the characteristic of low coherence. It is well known that the low-coherence or decorrelation areas are not always areas of change. As described in Section I, the coherence values may be also low in vegetation or low-CNR areas. However, in these areas, although decorrelation, it is unlikely to exist a large difference between the radar echoes of two repeat-pass SAR images if the features in these areas have not changed. In other words, the intensity of two images with identical collection geometries is approximate in these regions where no features

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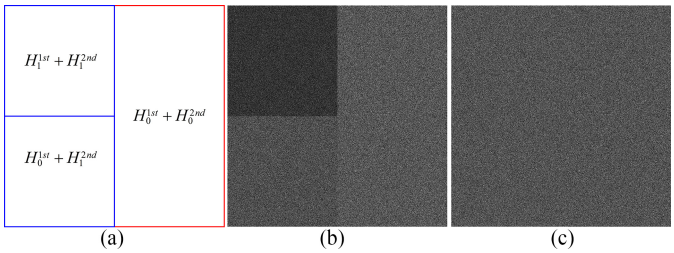

Fig. 2. Design frame and real components of the simulation SAR image pair. (a) Simulation design frame where the top-left patch is simulated using the test $H_1^{1st} + H_1^{2nd}$, the bottom-left patch using the test $H_0^{1st} + H_1^{2nd}$, and the right patch using the test $H_0^{1st} + H_0^{2nd}$. (b) and (c) Corresponding real components of two SAR images of the simulated pair, respectively.

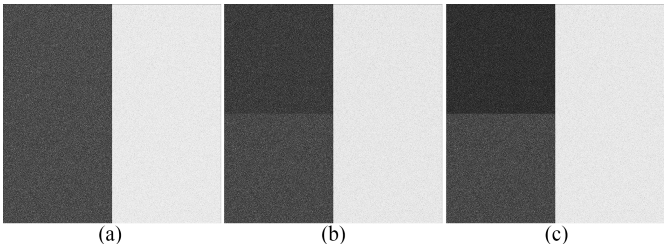

Fig. 3. SAR change detection images of the simulation image pair estimated using estimators. (a) $\hat{\gamma}_c$. (b) $\hat{\gamma}_b$. (c) $\hat{\gamma}_w$.

have changed. This situation is exactly in accord with the test of $H_0^{1st}+H_1^{2nd}$, i.e., $\sigma_f \approx \sigma_g$ and $\gamma < 1$. If the situations where the coherence is low and a significant intensity difference exists occur, it usually represents that a true temporal change has occurred between two data collections. At this point, the test $H_1^{1st} + H_1^{2nd}$ contains this situation. Consequently, in the case of low-coherence images, we expect that the estimator is capable of discriminating two tests, identifying the change decorrelation areas from low decorrelation areas, rather than just highlighting all low-coherence areas.

To intuitively display the effectiveness of estimators in low- and high-coherence regions, the simulated image pair consists of three parts: the 1000 × 1000 top-left simulated patch relied on test $H_1^{1st}+H_1^{2nd}$, the 1000 × 1000 bottom-left patch relied on $H_0^{1st} + H_1^{2nd}$, and the 1000 × 2000 right patch relied on $H_0^{1st}+H_0^{2nd}$ [see Fig. 2(a)]. To facilitate the understanding, two images of the pair are marked as I_f and I_g from here to the end. In the simulation, σ_g^2, i.e., the variance of I_g, is assumed to be 1.0. For the left patches with low coherence, the ratio R is set to different values: 0.5 ($\sigma_f \neq \sigma_g$) for the upper part and 0.9 ($\sigma_f \approx \sigma_g$) for the lower part. Also, for the right patch, the variance of the additive noise is set to half of that of I_g, that is, $\sigma_n^2 = 0.5\sigma_g^2$. In particular, the top-left image can be viewed as covering a scene of change due to the low coherence and the significant difference in intensity, while the bottom-left image as a scene characterized by low coherence and similar intensity. Obviously, the right patch corresponds to the image where no change has occurred. Since using the same simulation way, only real components of the simulated image pair are demonstrated in Fig. 2(b) and (c).

In practice, raw estimator outputs, that is, the coherence magnitude images, are often called SAR change detection images in CCD [2]. Fig. 3 shows the change detection images of the simulated image pairs estimated using estimators $\hat{\gamma}_c$, $\hat{\gamma}_b$, and $\hat{\gamma}_w$, respectively. Also, the corresponding fitting curves based on the histograms of three patches in these images are shown in Fig. 4. On the whole level, it is observed from Fig. 3 that the proposed estimator gets better performance than the other two estimators: the contrast between the three patches is most obvious as well as the boundaries. Furthermore, a comparative analysis of the results in Figs. 3 and 4 is mainly performed in low- and high-coherence regions, respectively, and the following conclusions can be drawn.

1) In low-coherence regions, the coherence image calculated by $\hat{\gamma}_c$ has no significant difference between the patches with different ratios of variance, which means that estimator $\hat{\gamma}_c$ fails to distinguish the change decorrelation from other low decorrelation. By contrast, $\hat{\gamma}_b$ is better than $\hat{\gamma}_c$, which exactly corresponds with [2]. Although improving, $\hat{\gamma}_b$ is still inferior to $\hat{\gamma}_w$, as the result of $\hat{\gamma}_w$ can provide a more obvious contrast of the coherence between two low-coherence patches. The conclusion can be further demonstrated by Fig. 4(a) and (b). It is observed that the new estimator $\hat{\gamma}_w$ can significantly stretch the coherence toward the smaller-value direction than $\hat{\gamma}_c$ and $\hat{\gamma}_b$ when the variance ratio is low. For a large value of R, this stretching effect is not obvious, and the coherence obtained using three estimators almost has the same values. Therefore, $\hat{\gamma}_w$ is very sensitive to the ratio R and is more effective than $\hat{\gamma}_c$ and $\hat{\gamma}_b$ for identifying change areas and other low-coherence areas.
2) In high-coherence regions, all estimators can get high coherence values: the mean values of three fitting curves are about 0.9. The coherence images estimated using three estimators have visually no obvious difference, and the fitting curves based on their histograms only have a slight difference.

From the abovementioned analysis in low- and high-coherence areas, we can conclude that the proposed estimator can yield the best change detection image among the three estimators in theory. In particular, the detection result of $\hat{\gamma}_w$ presents higher contrast than that of $\hat{\gamma}_c$ and $\hat{\gamma}_b$, because the weighted estimator can effectively lower coherence values in areas with low coherence and relatively large differences in intensity and provide similar coherence values as $\hat{\gamma}_c$ and $\hat{\gamma}_b$ in other areas. In interferometric SAR tasks for terrain generation, the larger the magnitude of the correlation coefficient, the better because the quality of interference fringes improves with the coherence value [9], [21]. Differently, the purpose of CCD is to detect changes that have occurred between two SAR image collects; thus, an optimal coherence image should increase the contrast between the changed and unchanged regions, i.e., restraining the information of the background and enhancing that of the changed areas to the greatest extent [22]. Therefore, the weighted estimator $\hat{\gamma}_w$ is theoretically more suitable for CCD tasks, and its applicability in real SAR image pairs will be proved in Section V.

V. Real Data Example

A. Available Image Pairs

To further assess the effectiveness of the proposed estimator and the applicability in satellite SAR images, three pairs of

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TABLE I

AVAILABLE IMAGE PAIRS AND THEIR INFORMATION. TSX IS THE ABBREVIATION FOR TERRASAR-X AND TDX FOR TANDEM-X. RES IS THE ABBREVIATION FOR RESOLUTION AND IM FOR IMAGING MODE

Set	Sensor	Angle	Date	Res	IM	Size
1	TSX	42.4°	02/06/2013	1m	SpotLight	1000×1000
	TSX	42.4°	05/16/2013	1m	SpotLight	1000×1000
2	TSX	41.1°	05/24/2015	3m	StripMap	800×800
	TSX	41.1°	10/25/2015	3m	StripMap	800×800
3	TDX	41.1°	07/07/2015	3m	StripMap	4000×3000
	TSX	41.1°	07/29/2015	3m	StripMap	4000×3000

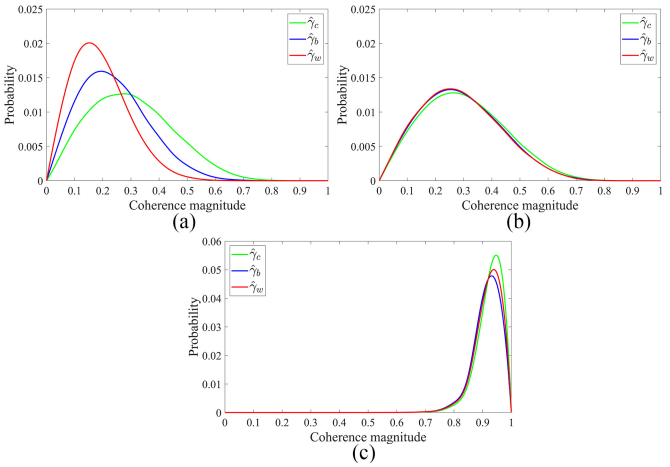

Fig. 4. Fitting probability curves obtained from the histograms in (a) top-left patch, (b) bottom-left patch, and (c) right patch of the change detection images in Fig. 3.

TerraSAR-X (TSX)/TanDEM-X (TDX) images in different IMs are used in the experiments. These pairs are selected as test sites for detecting changes in different areas: dataset 1 is utilized to detect ice changes in water areas, dataset 2 to detect changes in building areas, and dataset 3 to identify vegetation changes. The available sets of image pairs and their detailed information are demonstrated in Table I.

Fig. 5 shows the SAR image pairs used in the experiments and their corresponding ground truths. As CCD can identify subtle change, all SAR images are without any multilooking processing to keep this advantage. Moreover, as possessing the characteristic of a large dynamic range [23], SAR images used in this article are 16 bits to keep more details of radar echo. One point to be mentioned here is that the ground truth (reference image) was determined by visual inspection of two SAR images integrating prior information and recorded as a binary image.

B. Experiment Parameters

For all methods, the window size of coherence estimation is set to 3×3. Besides, the variances of the thermal additive noise of two SAR images are also needed for the estimator $\hat{\gamma}_\alpha$. As considered to be constant throughout the scene [24], this noise can be measured in areas of low average backscatter coefficient [25]. In this article, the noise values are measured in a manually selected shadow area following the procedure introduced in [8]. For the proposed estimator, a practical question involves the determination of R at each pixel. As σ^2 is the mean power, this suggests that the correct approach, given L-independent measurements, is to average the measurements

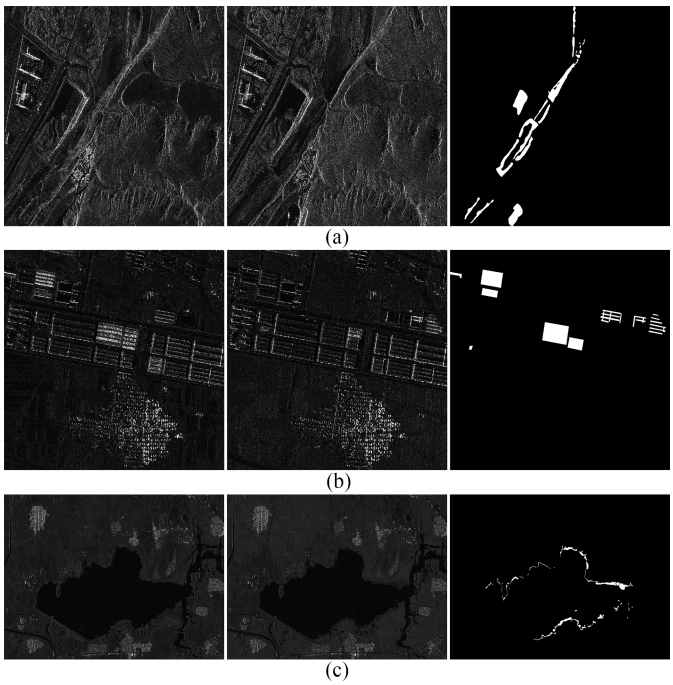

Fig. 5. Experimental SAR image pairs and their corresponding ground truths. (a) Dataset 1. (b) Dataset 2. (c) Dataset 3.

in intensity [25]. Therefore, to reduce the effect of speckles, the neighborhood averaging window size to calculate R is set to 3×3.

C. Results and Analysis

This section is divided into two parts. A detailed comparison is first investigated between the results of the weighted estimator and the other three typical coherence estimators, i.e., the mentioned $\hat{\gamma}_c$, $\hat{\gamma}_b$, and $\hat{\gamma}_\alpha$ in Section II. Then, a brief comparison between the mean-ratio image and change images obtained using estimators $\hat{\gamma}_b$ and $\hat{\gamma}_w$ is performed on dataset 1 to evaluate the new estimator. All algorithms are developed with MATLAB R2019b.

1) Performance Comparison With Other Estimators: In Figs. 6–8, we show the raw outputs of $\hat{\gamma}_c$, $\hat{\gamma}_b$, $\hat{\gamma}_\alpha$, and $\hat{\gamma}_w$, respectively, prior to thresholding. Note that, in three detection images of $\hat{\gamma}_c$, the estimator can provide low values in both change areas and other decorrelation areas, such as water and vegetation areas, but fails to distinguish these two types of areas from each other. This is exactly consistent with the conclusion on the simulated SAR image pair. By contrast, the detection images in Figs. 6–8 with the estimator $\hat{\gamma}_b$ are much better than $\hat{\gamma}_c$, differentiating the changes and presenting significantly darker pixels in change areas. Also, $\hat{\gamma}_\alpha$ can be essentially seen as a modification of $\hat{\gamma}_b$, capable of separating low-CNR areas from other decorrelation areas. This capability makes the change detection images of $\hat{\gamma}_\alpha$ in low-CNR areas provide high coherence values; for example, the water surfaces in the middle of Fig. 8(c) fill with many bright pixels. Besides, it can be seen from Figs. 6–8 that the overall brightness of the detection results of $\hat{\gamma}_\alpha$ has slightly improved compared with $\hat{\gamma}_b$ except in change areas where slight darker pixels are filled with. However, the performance of these two esti-

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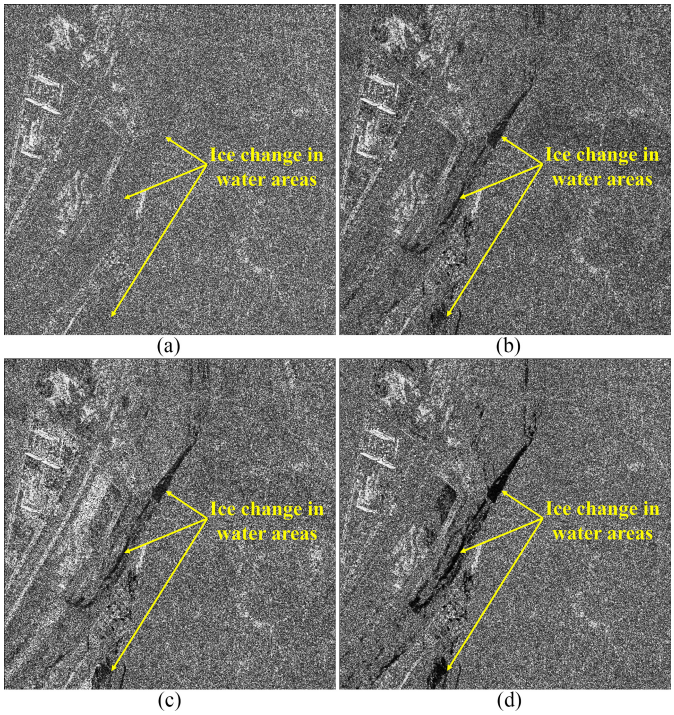

Fig. 6. SAR change detection images obtained by different coherence estimators for dataset 1. (a) $\hat{\gamma}_c$. (b) $\hat{\gamma}_b$. (c) $\hat{\gamma}_\alpha$. (d) Proposed $\hat{\gamma}_w$.

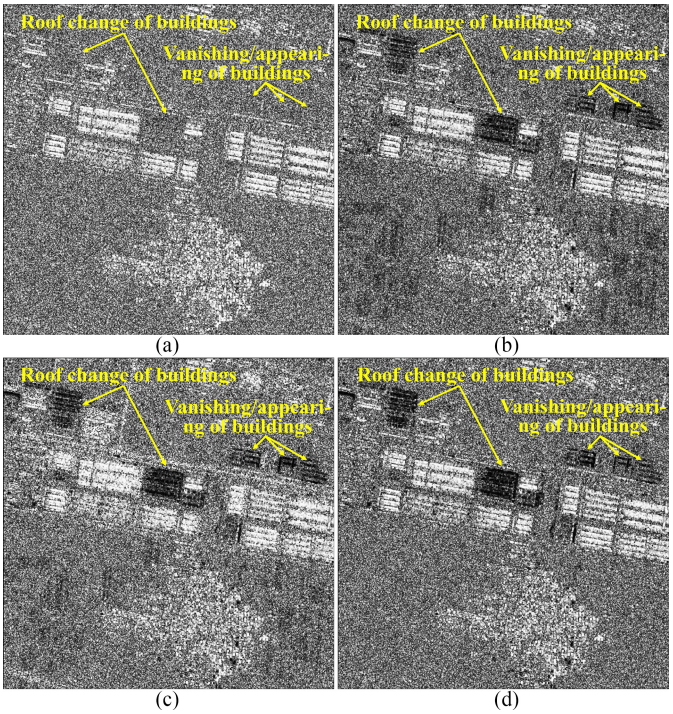

Fig. 7. SAR change detection images obtained by different coherence estimators for dataset 2. (a) $\hat{\gamma}_c$. (b) $\hat{\gamma}_b$. (c) $\hat{\gamma}_\alpha$. (d) Proposed $\hat{\gamma}_w$.

mators is still relatively limited, and change regions are not effectively highlighted, especially the boundaries of change and no-change areas. The boundaries are fuzzy and difficult to determine, which is unfavorable for obtaining fine change maps (see close-up images in Fig. 9). Comparatively, the results of the weighted estimator better highlight the changes by significantly lowering the coherence values in change areas and show a higher contrast between changed and unchanged areas.

Smaller portions of change areas in Figs. 6–8 are shown in Fig. 9 to clearly display the difference between the four estimators. It is observed that the close-up images of $\hat{\gamma}_c$ cannot effectively highlight change information (see all left figures in Fig. 9). Therefore, a comparative analysis is mainly performed among $\hat{\gamma}_b$, $\hat{\gamma}_\alpha$, and $\hat{\gamma}_w$. Fig. 9(a) and (b) shows the close-up images extracted from Fig. 6 in deterministic ice change areas. Notice that $\hat{\gamma}_b$ can roughly identify areas where the ice surface changes, but the boundaries of ice-changed regions are very unclear. The result from $\hat{\gamma}_\alpha$ is slightly better than $\hat{\gamma}_b$: most ice-change regions fill with relatively darker pixels, and the boundaries are somewhat more clear. Although improving, it is still not easy to identify the boundaries between changed and unchanged areas, thus bringing difficulties in the subsequent thresholding processing to get an optimal change map. By contrast, the images obtained using the weighted estimator significantly improve the results [see right figures of Fig. 9(a) and (b)]. The results from $\hat{\gamma}_w$ fill with much darker pixels than results from $\hat{\gamma}_b$ and $\hat{\gamma}_\alpha$ in the ice change regions, indicating easier identification of the boundary and less confusion with other low-coherence areas.

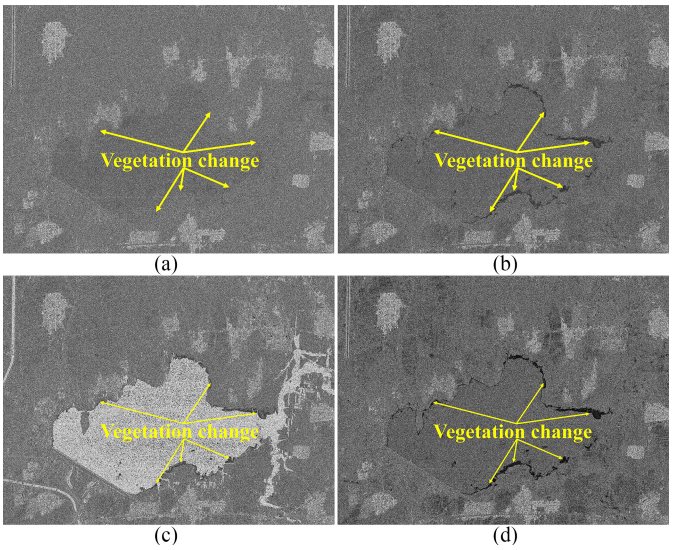

Fig. 8. SAR change detection images obtained by different coherence estimators for dataset 3. (a) $\hat{\gamma}_c$. (b) $\hat{\gamma}_b$. (c) $\hat{\gamma}_\alpha$. (d) Proposed $\hat{\gamma}_w$.

Fig. 9(c) and (d) shows the zoomed-in images extracted from Fig. 7 in areas where buildings appear and roofs change, respectively. It can be observed that the performance of $\hat{\gamma}_w$ is the best, $\hat{\gamma}_\alpha$ the next, and $\hat{\gamma}_b$ the last, which is consistent with the abovementioned conclusion. Here, we also observe that the detection results are not continuous building shapes, which is closely related to the ability of CCD in the detection of small-scale changes [2]. Consequently, many change details in buildings have been preserved. Besides, the resolution of this image pair is 3 m, relatively low for detecting building

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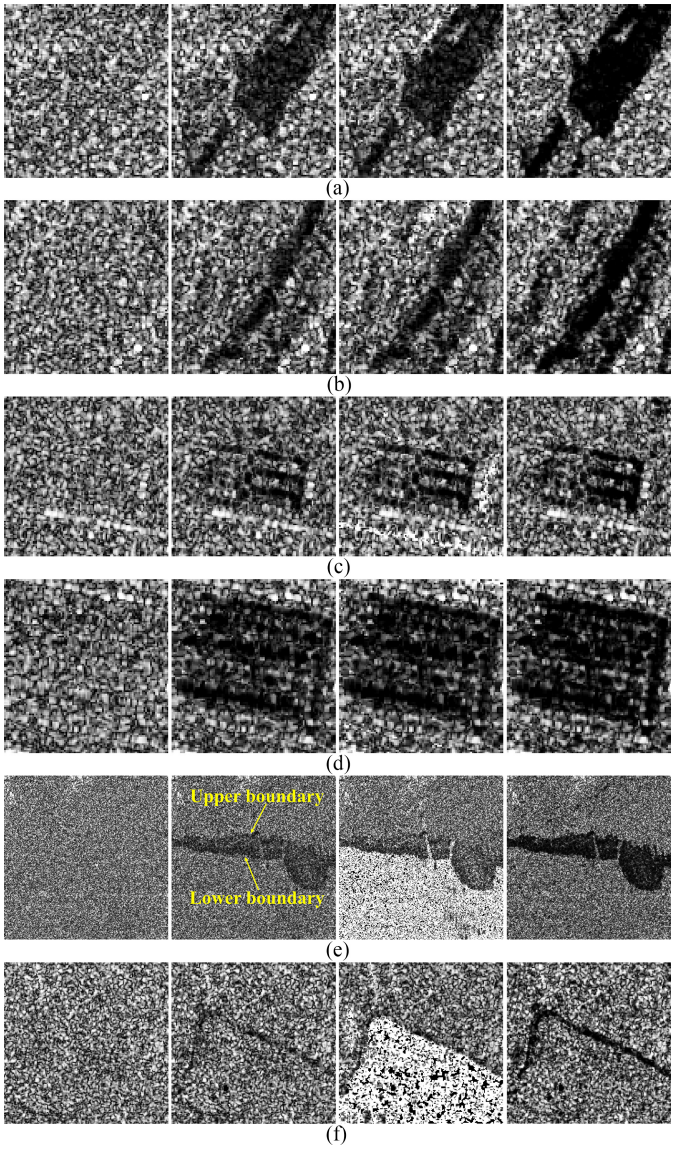

Fig. 9. Corresponding close-up images of part of change areas, extracted from Figs. 6–8, respectively. (a) and (b) Ice change areas extracted from Fig. 6. (c) and (d) Areas where buildings appear and roofs change extracted from Fig. 7. (e) and (f) Vegetation change areas extracted from Fig. 8.

change, so the detection result of buildings is not as good as that of features in large-scale homogenous regions (e.g., the results shown in Figs. 6 and 8).

Part details of Fig. 8 are demonstrated in Fig. 9(e) and (f). To facilitate the following explanation, the boundary of the change area is divided into two parts: the upper boundary between changed and unchanged areas of vegetation and the lower boundary between the changed vegetation area and the water area [see Fig. 9(e)]. It is found that the result of $\hat{\gamma}_\alpha$ is inferior to $\hat{\gamma}_b$. As mentioned, $\hat{\gamma}_\alpha$ provides high values in the low-CNR water area, making its lower boundary clearer than that of $\hat{\gamma}_b$. However, the upper boundary becomes very blurry, making it difficult to be determined [see the third images from the left in Fig. 9(e) and (f)]. In all cases, the proposed estimator $\hat{\gamma}_w$ achieves the best detection results with showing the highest contrast between the changed and unchanged areas. Overall, the contrast is emphasized in the detection results of $\hat{\gamma}_w$, making the change even sharper.

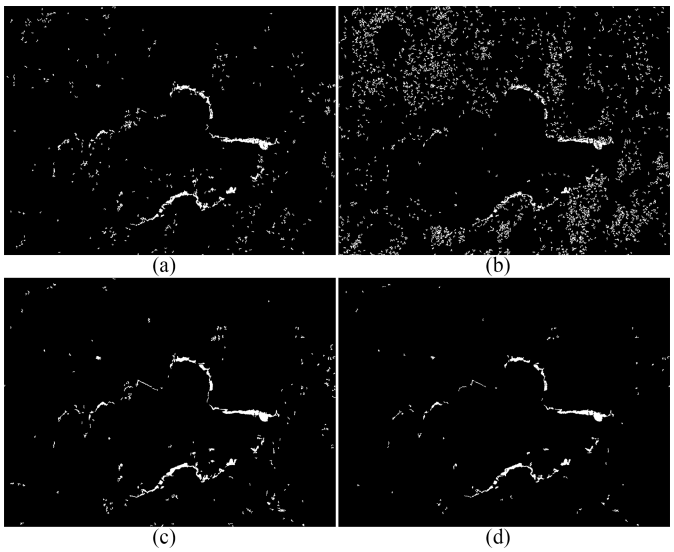

Fig. 10. SAR change detection maps obtained from change detection images of different estimators for dataset 3. (a) $\hat{\gamma}_b$. (b) $\hat{\gamma}_\alpha$. (c) $\hat{R}_m$. (d) Proposed $\hat{\gamma}_w$.

Before, we have made a detailed analysis of the raw estimator outputs. Because theoretically proved the effectiveness of the proposed estimator in Section IV and not studied algorithms to obtain the final change map, such as thresholding methods, in this article, we only selected one dataset for simple processings to evaluate the new estimator numerically. In this test, dataset 3 was selected, because it covers a large scene and has obvious differences in vegetation growth, which will have a significant impact on getting the final change map. Since $\hat{\gamma}_c$ fails to identify change regions from other decorrelation regions, a comparison is performed between the new $\hat{\gamma}_w$ and estimators $\hat{\gamma}_b$ and $\hat{\gamma}_\alpha$. In addition, an extension comparison between $\hat{\gamma}_w$ and the mean-ratio operator is made in Section V-C2; therefore, we also present the detection results of the mean-ratio operator. Fig. 10 demonstrates the SAR change maps of four different methods for dataset 3. One point to be mentioned here is that the final change map is obtained using the processes of the density slice and sieve in the Environment for Visualizing Images (ENVI). In the density slice process, the segmentation threshold is determined by the artificial experience. This threshold is easy to obtain, because the statistics provide low values in change areas. Also, the sieve threshold is set to 20 pixels, which means that the segmentation block smaller than this threshold would be ignored.

Compared change maps of $\hat{\gamma}_b$, $\hat{\gamma}_\alpha$, and $\hat{\gamma}_w$ in Fig. 10 with the ground truth in Fig. 5(c), it can be found that the proposed $\hat{\gamma}_w$ gives the best performance. The false alarms of $\hat{\gamma}_w$ are less than that of $\hat{\gamma}_b$ and $\hat{\gamma}_\alpha$, especially in vegetation regions. In addition, we found that false alarms of $\hat{\gamma}_\alpha$ are much more in vegetation areas but fewer in the water area than that of $\hat{\gamma}_b$. The reasons are as follows. First, the coherence magnitude using $\hat{\gamma}_\alpha$ is significantly improved in water areas, leading to an obvious reduction of false alarms in these areas. Second, since low vegetation also presents somewhat low-CNR properties, the coherence magnitude estimated by $\hat{\gamma}_\alpha$ in vegetation areas will increase as well. This increase, coupled with the purpose of dataset 3 identifying vegetation changes, makes the coherence

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT CHANGE STATISTICS FOR DATASET 3

Statistics	Pc	R	OA	Kappa
$\hat{\gamma}_b$	0.4270	0.8593	0.9883	0.5652
$\hat{\gamma}_\alpha$	0.1171	0.6600	0.9521	0.1862
$\hat{R}_m$	0.5513	0.9464	0.9925	0.6932
$\hat{\gamma}_w$	0.7522	0.8681	0.9962	0.8041

magnitude of change areas also increase, which is also the main reason that the upper boundary of the SAR change detection image obtained by $\hat{\gamma}_\alpha$ becomes blurry in Fig. 9(e). Also, the proposed $\hat{\gamma}_w$ can comparatively effectively filter out false alarms in water and vegetation regions by reducing the coherence magnitude in change areas and increasing the contrast between changed and unchanged areas, thus giving better performance.

Besides, to quantitatively show the performance of different change statistics, we also present the corresponding quantitative precision results of four change statistics, as shown in Table II. Notice that the following four criteria are utilized. The first one is the ratio of the correctly detected change pixels to all detected change pixels, referred to as Pc. The second is the recall ratio (R), the ratio of the correctly detected change pixels to all true change pixels. In addition, two comprehensive measures, overall accuracy (OA) and Kappa coefficient [26], are employed. OA is the ratio of all correctly detected pixels, including changed and unchanged, to all pixels of the image. Also, the Kappa coefficient is calculated from the confusion matrix. The quantitative results of $\hat{\gamma}_b$, $\hat{\gamma}_\alpha$ and proposed $\hat{\gamma}_w$ in Table II further demonstrate the best performance of the proposed estimator. Compared with the other two estimators, all precision measures of $\hat{\gamma}_w$ have been improved obviously.

2) Extension Comparison: In addition to the detailed comparison between the weighted estimator and several typical coherence estimators, a brief comparison is also performed between the mean-ratio image and raw outputs from estimators $\hat{\gamma}_b$ and $\hat{\gamma}_w$. There are two reasons for this comparison: one is the utilization of the mean-ratio parameter in the calculation of $\hat{\gamma}_w$ [see (15)], and the other is $\hat{\gamma}_w$ that can be seen as a weighted improvement of $\hat{\gamma}_b$. Actually, the mean-ratio image is often called the difference image (DI) in noncoherent intensity-based change detection methods. Here, for consistency with the definition of raw estimator outputs, the mean-ratio image is referred to as the SAR change detection image as well. In this experiment, the mean-ratio image is generated from the following mean ratio operator as:

$$\hat{R}_m(p) = \min\left(\frac{\bar{I}_1(p)}{\bar{I}_2(p)}, \frac{\bar{I}_2(p)}{\bar{I}_1(p)}\right) \tag{26}$$

where $\bar{I}_1$ and $\bar{I}_2$ are the mean intensities estimated from the local neighborhood around the pixel p in two spatial-registered SAR images, respectively. Different from the definition in [27], (26) is not subtracted from 1; thus, in change areas, low mean-ratio values can be obtained, keeping the same trend as the coherence estimators. Also, the averaging window

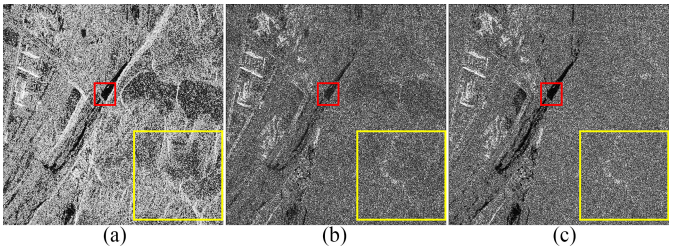

Fig. 11. SAR change detection images for dataset 1. (a) $\hat{R}_m$ with a window size of 3 × 3. (b) $\hat{\gamma}_b$. (c) Proposed $\hat{\gamma}_w$.

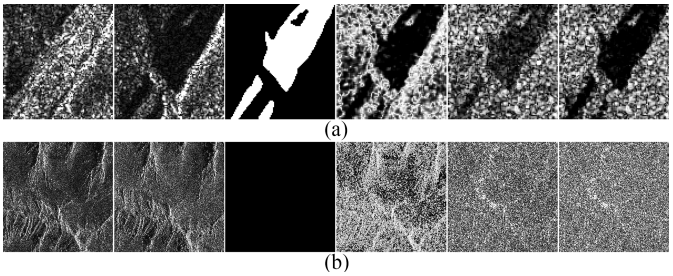

Fig. 12. Close-up images of dataset 1 in (a) changed area and (b) unchanged area marked with red and yellow rectangles in Fig. 11, respectively. The images from left to right correspond to two partial SAR images of dataset 1, corresponding ground truths, and change detection images obtained from $\hat{R}_m$, $\hat{\gamma}_b$, and $\hat{\gamma}_w$.

to obtain the mean-ratio image is set to 3 × 3, the same as the size to calculate R in the estimation of $\hat{\gamma}_w$.

Fig. 11 shows the raw SAR change detection images of these three statistics for dataset 1. It can be seen from Fig. 11(a) that $\hat{R}_m$ provides low values in the ice-change areas; however, it is very sensitive to intensity changes as an intensity-based statistical measure, making many unchanged regions also with dark pixels. In Fig. 5(a), we observe that the right scene of dataset 1 covers a mountain area. Due to the different acquisition times, natural processes such as vegetation growth can inevitably bring about intensity differences between two SAR images. In most cases, we are not interested in these differences, which, however, are highlighted in the mean-ratio image [see Fig. 11(a)], bringing difficulties in extracting the change information we want. By contrast, these differences are comparatively weakened in change detection images of coherence estimators [see Fig. 11(b) and (c)]. In addition, the sensibility of $\hat{R}_m$ in intensity changes is exactly the reason the boundary of change areas in the mean-ratio image is comparatively clearer than that in the detection image of $\hat{\gamma}_b$. Note that the result of $\hat{\gamma}_w$ nicely combines the advantages of $\hat{R}_m$ and $\hat{\gamma}_b$, enhancing the information in change areas and restraining the information in unchanged areas to great extent. To further illustrate this advantage, two close-up images in changed and unchanged areas are shown in Fig. 12. Notice in Fig. 12(a) that, in change areas, the result of $\hat{\gamma}_w$ retains the advantage of $\hat{R}_m$, displaying a clear boundary, while in the unchanged area in Fig. 12(b), it keeps the advantage of $\hat{\gamma}_b$, not so sensitive to intensity changes. In addition, from the SAR change detection maps of $\hat{\gamma}_b$, $\hat{R}_m$ and proposed

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$\hat{\gamma}_w$ in Fig. 10 and their corresponding quantitative precision results in Table II, the weighted estimator can effectively improve the results from $\hat{\gamma}_b$ and $\hat{R}_m$. From the abovementioned comparison, we can conclude that the new estimator can achieve an effective combination of the two statistics and improve detection precision, by adding weight parameters related to the mean ratio on estimator $\hat{\gamma}_b$.

VI. Conclusion

In this article, a weighted coherence estimator for SAR CCD has been proposed. This estimator is derived based on the statistical characteristics of SAR images using the ML principle. The estimator has been shown to have a better change detection performance over the current common estimators and can effectively increase the contrast between the changed and unchanged regions by significantly lowering the coherence values in areas with low coherence and relatively large differences in intensity. Besides, as the weight parameters are closely related to the ratio change metric, its introduction not only makes the estimator no longer need to satisfy the equal-variance assumption but also combines the advantages of coherent and noncoherent intensity-based algorithms to a certain extent.

The experimental results on simulated and real SAR image pairs verify the effectiveness of the proposed estimator for SAR change detection. Moreover, the applicability of the proposed estimator in satellite SAR images has also been validated. In future works, we are going to focus on algorithms to get the final change map from the change detection image of the proposed estimator.

Appendix
Derivation of $\hat{\gamma}_{ml}$

Given two spatially registered complex SAR datasets f and g of N pixels, a joint data vector $\boldsymbol{X} = [\boldsymbol{f}, \boldsymbol{g}]^T \in \mathbb{C}^{N\times 2}$ can be formed. As the mutual independence of the N sample pairs of observations $\boldsymbol{X}_k = [f_k, g_k], k = 1, 2, \ldots, N$, a joint conditional density function can be expressed as

$$p(\boldsymbol{X}|\gamma, \phi) = \frac{1}{\pi^{2N}|\boldsymbol{Q}|^N} \exp\Big(-\sum \boldsymbol{X}_k^H \boldsymbol{Q}^{-1}\boldsymbol{X}_k\Big) \tag{A1}$$

where the superscript H is the complex conjugate transpose operation, and $|\boldsymbol{Q}|$ is the determinant of the covariance matrix $\boldsymbol{Q}$ given by

$$\boldsymbol{Q} = \mathbb{E}(\boldsymbol{X}\boldsymbol{X}^H) = \begin{bmatrix} \sigma_f^2 & \gamma\,\sigma_f\sigma_g e^{j\phi} \\ \gamma\,\sigma_f\sigma_g e^{-j\phi} & \sigma_g^2 \end{bmatrix} \tag{A2}$$

where $\sigma_f^2 = \mathbb{E}(|\boldsymbol{f}|^2)$, $\sigma_g^2 = \mathbb{E}(|\boldsymbol{g}|^2)$, γ is the magnitude of the complex cross-correlation coefficient, and ϕ denotes the phase difference that exists between f and g.

The ML estimate of γ is the value of γ maximizing (A1) or, equivalently, the log of (A1), that is,

$$\ln p(\boldsymbol{X}|\gamma, \phi) = -2N\ln\pi - N\ln|\boldsymbol{Q}| - \sum \boldsymbol{X}_k^H \boldsymbol{Q}^{-1}\boldsymbol{X}_k. \tag{A3}$$

Taking the partial derivative of (A3) concerning γ and setting the result to zero give

$$-\frac{N}{|\boldsymbol{Q}|}\frac{\partial(|\boldsymbol{Q}|)}{\partial\gamma} - \sum \boldsymbol{X}_k^H \frac{\partial(\boldsymbol{Q}^{-1})}{\partial\gamma}\boldsymbol{X}_k = 0 \tag{A4}$$

where $|\boldsymbol{Q}|$ is the determinant of $\boldsymbol{Q}$, and $\boldsymbol{Q}^{-1}$ is the inverse matrix expressed as

$$|\boldsymbol{Q}| = (1-\gamma^2)\sigma_f^2\sigma_g^2 \tag{A5}$$

$$\boldsymbol{Q}^{-1} = \frac{1}{1-\gamma^2}\begin{bmatrix} \frac{1}{\sigma_f^2} & -\frac{\gamma e^{j\phi}}{\sigma_f\sigma_g} \\ -\frac{\gamma e^{-j\phi}}{\sigma_f\sigma_g} & \frac{1}{\sigma_g^2} \end{bmatrix}. \tag{A6}$$

The partial derivation can be written as

$$\frac{\partial|\boldsymbol{Q}|}{\partial\gamma} = -2\gamma\,\sigma_f^2\sigma_g^2 \tag{A7}$$

$$\begin{aligned}\frac{\partial(\boldsymbol{Q}^{-1})}{\partial\gamma} &= \frac{2\gamma}{(1-\gamma^2)^2}\begin{bmatrix} \frac{1}{\sigma_f^2} & -\frac{\gamma e^{j\phi}}{\sigma_f\sigma_g} \\ -\frac{\gamma e^{-j\phi}}{\sigma_f\sigma_g} & \frac{1}{\sigma_g^2} \end{bmatrix} \\ &\quad + \frac{1}{1-\gamma^2}\begin{bmatrix} 0 & -\frac{e^{j\phi}}{\sigma_f\sigma_g} \\ -\frac{e^{-j\phi}}{\sigma_f\sigma_g} & 0 \end{bmatrix} \\ &= \frac{1}{(1-\gamma^2)^2}\begin{bmatrix} \frac{2\gamma}{\sigma_f^2} & \frac{-(1+\gamma^2)}{\sigma_f\sigma_g}e^{j\phi} \\ \frac{-(1+\gamma^2)}{\sigma_f\sigma_g}e^{-j\phi} & \frac{2\gamma}{\sigma_g^2} \end{bmatrix}.\end{aligned} \tag{A8}$$

Substituting the definition of $\boldsymbol{X}_k$ and (A5)–(A8) into (A4), we have

$$\begin{aligned}&\frac{\partial}{\partial\gamma}\ln p(\boldsymbol{X}|\gamma,\phi) \\ &\quad = \frac{2N\gamma}{1-\gamma^2} - \frac{1}{(1-\gamma^2)^2} \\ &\qquad \times\left[2\gamma\left(\frac{\sum f_k^* f_k}{\sigma_f^2} + \frac{\sum g_k^* g_k}{\sigma_g^2}\right) - \frac{(1+\gamma^2)}{\sigma_f\sigma_g}\right. \\ &\qquad \left.\times\left(e^{-j\phi}\sum f_k g_k^* + e^{j\phi}\sum f_k^* g_k\right)\right] = 0.\end{aligned} \tag{A9}$$

Using the following reasonable approximations:

$$\sigma_f^2 \approx \frac{1}{N}\sum f_k^* f_k \tag{A10}$$

$$\sigma_g^2 \approx \frac{1}{N}\sum g_k^* g_k \tag{A11}$$

we obtain

$$\frac{\sum f_k^* f_k}{\sigma_f^2} + \frac{\sum g_k^* g_k}{\sigma_g^2} \approx 2N. \tag{A12}$$

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Substituting (A12) into (A9), then dividing through by $1+\gamma^2$ and simplifying, we get

$$\gamma = \frac{e^{-j\phi}\sum f_k g_k^* + e^{j\phi}\sum f_k^* g_k}{\frac{\sigma_g}{\sigma_f}\sum f_k^* f_k + \frac{\sigma_f}{\sigma_g}\sum g_k^* g_k}. \tag{A13}$$

The ML estimate of the phase difference ϕ (see [9]) is

$$\hat{\phi}_{ML} = \angle \sum f_k g_k^* \tag{A14}$$

where the operator $\angle$ is the phase of the complex quantity.

Therefore

$$e^{j\hat{\phi}_{ML}} = \frac{\sum f_k g_k^*}{\left|\sum f_k g_k^*\right|}. \tag{A15}$$

Also

$$\begin{aligned} e^{j\hat{\phi}_{ML}}\sum f_k^* g_k &= \frac{\sum f_k g_k^*}{\left|\sum f_k g_k^*\right|}\sum f_k^* g_k = \frac{\left(\sum f_k^* g_k\right)^*}{\left|\sum f_k^* g_k\right|}\sum f_k^* g_k \\ &= \left|\sum f_k^* g_k\right|. \end{aligned} \tag{A16}$$

Similarly

$$e^{-j\hat{\phi}_{ML}}\sum f_k g_k^* = \left|\sum f_k^* g_k\right|. \tag{A17}$$

Substituting (A16) and (A17) into (A13) and setting the ratio of the true variance as $R = \sigma_f^2/\sigma_g^2$, we have

$$\hat{\gamma}_{ml} = \frac{2\left|\sum f_k^* g_k\right|}{\sum f_k^* f_k \Big/ \sqrt{R} + \sqrt{R}\sum g_k^* g_k}. \tag{A18}$$

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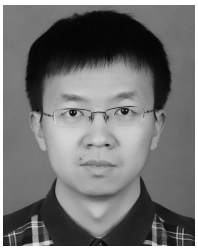

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